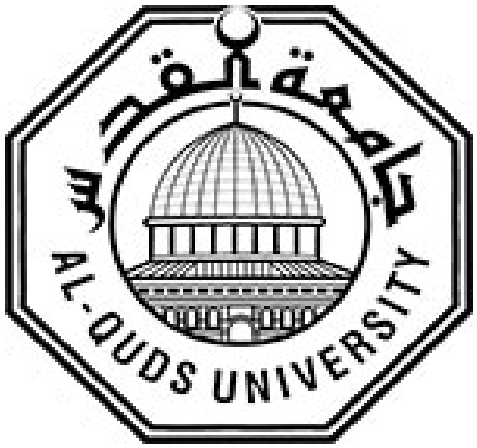


AL – QUDS UNIVERSITY

FACULTY OF ENGINEERING

DEPARTMENT OF
COMPUTER ENGINEERING

PROGRAM OF STUDY FOR THE B.Sc. DEGREE

IN

COMPUTER ENGINEERING

MARCH 2010

AL - QUDS UNIVERSITY
FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER ENGINEERING

PROGRAM OF STUDY FOR THE B.Sc. DEGREE IN
COMPUTER ENGINEERING
March 2010

Admission Requirements:

New students can apply for admission to this program upon obtaining a minimum 80% average grade in the Tawjihi-Scientific Stream Certificate or equivalent. Final admission is, however, granted upon satisfying the Tawjihi grade standards and other general requirements set by the Faculty in conjunction with the University Admission Office in the beginning of each academic year.

Program Objectives:

The educational objectives of the Computer Engineering Department are to turn out graduates who:

1. Are competent engineers with solid foundations in Computers.
2. Are problem-solvers, innovators, and can handle the rapid pace of modern computer technology.
3. Can function effectively in professional environment, have the necessary communication and leadership skills, and adhere to standard codes of ethics.
4. Can pursue graduate studies and carry out research in advanced topics and specializations.
5. Produce gruates with the nessary background and technical skills to work professionally in one or more of the following areas: computer hardware and software design, computer-based systems, computer network design, system integration, electronic design automation.

Program Intended Learning Outcomes (ILO's)

The ILO's of the Computer Engineering program adhere to the American Accreditation Board for Engineering and Technology (ABET). Thus, upon completion of this program, students will have:

a) An ability to apply knowledge of mathematics, science, and engineering.
b) An ability to design and conduct experiments, to analyze and interpret data.
c) An ability to design a system, component, or process to meet desired needs.
d) An ability to function on multidisciplinary teams.
e) An ability to identify, formulate, and solve engineering problems.
f) An understanding of professional and ethical responsibility.
g) An ability to communicate effectively.
h) The broad education necessary to understand the impact of engineering solutions in a global and societal context.
i) A recognition of the need for, and an ability to engage in life-long learning.
j) A knowledge of contemporary issues.
k) An ability to use the techniques, skills, and modern engineering tools necessary for engineering practice.

Structure and Contents:

The B.Sc. degree in Computer Engineering is awarded upon completing a total of 166 credit hours according to the following requirements:

No.	Requirements	Credit Hours of Required Courses			Total
		Compulsory	Elective	Free	
I	University Requirements	18	6	-	24
II	Faculty Requirements	44	3	6	53
III	Program Requirements	82	9	0	91
				Grand Total	168

I- University Requirements (Total 24 credits)

Students must successfully complete:

- 18 credit hours of compulsory courses from **Table I-a**.
- 6 credit hours of elective courses consisting of either two foreign language courses (introductory + intermediate) form **Table I-b** or any two courses from **Table I-c**.

Compulsory Courses (18 credits) Table I-a

Course No.	Course Name	Credits
0400101	Arabic Language Skills	2
0400108	English Language Skills 1	2
0400109	English Language Skills 2	2
0400120	Jerusalem Through History	2
0400121	Nature and Environment of Palestine	2
0400122	Islamic Culture	2
0400124	Logical Thinking	3
0400128	International Civilizations	3

Elective Courses (6 credits)

Table I-b

Course No.	Course Name	Credits
0400111	Hebrew Language I	3
0400112	Hebrew Language II	3
0400113	French Language I	3
0400114	French Language II	3
0400115	German Language I	3
0400116	German Language II	3
0400117	Spanish Language I	3
0400118	Spanish Language II	3
0400119	Turkish Language I	3
0400129	Turkish Language II	3
0400146	Italian Language I	3
0400147	Italian Language II	3

Table I-c

Course No.	Course Name	Credits
0300142	Science and Life	3
0303100	Internet for Special Purposes	3
0305100	Introduction to Public Health and Environment	3
0400130	Fine Arts	3
0400131	Introduction to Music	3
0400132	Literary Appreciation	3
0400133	Men and Women in the Human Society	3
0400141	Issues in Modern and Contemporary Arab Thinking	3
0400148	Special Topics	3
0400149	Islam and Contemporary Issues	3
0403131	Communication Skills	3
0409135	History and Philosophy of Science	3
0500140	Democracy, Human Rights & International Humanitarian Law	3
0500143	Solving Conflicts by Peaceful Means	3
0500145	Law and Society	3

II- Faculty Requirements (Total 53 credits)

Students must successfully complete:

- 44 credit hours of compulsory courses from **Table II-a**.
- 3 credit hours elective course from **Table II-b**.
- 6 credit hours of free courses.

Compulsory Courses(44 credits) Table II-a

Course No.	Course Name	Credits		Prerequisites /Corequisites	ILO's Covered
		Th	Lab		
0302103	General Physics I for Engineers I	3	-	---	a
0302104	General Physics II for Engineers II	3	-	P-0302103	a
0302113	General Physics I Laboratory	-	1	C-0302103	a,b,d,g
0302114	General Physics II Laboratory	-	1	C-0302104	a,b,d,g
0306101	Calculus I	3	-	---	a
0306102	Calculus II	3	-	P-0306101	a
0700103	Engineering Drawing	-	1	---	a,b,d,e,g,k
0700209	Technical Writing & Communication Skills I	3	-	P-0400109	a,d,g,h
0700231	Engineering Mathematics	3	-	P-0306102	a,e,k
0700233	Engineering Mathematics Laboratory	-	1	C-0700231	a,b,d,e,g,k
0700306	Thermodynamics	3	-	P-0700231	a,b,c,h,k
0700381	Engineering Management	3	-	3rd year +	a,d,g,i,k
0701211	Introduction to Circuit Theory & Electronics	3	-	P-0302104	a,e,k
0701213	Introduction to Circuit Theory & Electronics Laboratory	-	1	C-0701211	a,b,d,g,k
0702111	Introduction to Computer Engineering	3	-	---	a,d,e,f,h,j
0702112	Introduction to Computer Engineering Lab.	-	1	C-0702111	a,d,e,f,h,j
0702113	Programming Fundamentals for Engineers	3	-	P-0702111	a,b,c,e-k
0702114	Programming Fundamentals for Engineers Laboratory	-	1	C-0702113	a,b,c,e-k
0703201	Introduction to Materials Engineering	3	-	P-0302103	a,h,i,j,k
0703203	Introduction to Materials Engineering Lab.	-	1	C-0703201	b,c,k

Elective Courses (3 credits) Table II-b

Course No.	Course Name	Credits		Prerequisites	ILO's Covered
		Th	Lab		
0700382	Engineering Economy	3	-	3rd year +	a,e,i
0700383	Quality Control	3	-	3rd year +	a,b,i,k
0700385	E Business	3	-	3rd year +	a,c,d,e,f,h,g

Free Courses (6 credits)

Students must select a total of 6 credit hours of free courses, providing they obtain the consent of the Faculty.

III- Program Requirements (Total 91 credits)

Students must successfully complete:

- 82 credit hours of compulsory courses from **Table III-a**.
- 9 credit hours of elective courses from **Table III-b**.
- At least 135 field training hours.

A- Compulsory Courses (82 credits) Table III-a

Course No.	Course Name	Credits		Prerequisites /Corequisites	ILOs Covered
		Th	Lab		
0701212	Electric Circuits	3	-	P-0701211	a,e,k
0701214	Electric Circuits Lab.	-	1	C-0701212	a,b,d,g,k
0701222	Electronics	3	-	P-0701211	a,c,e,k
0701224	Electronics Lab.	-	1	C-0701222	a,b,d,g,k
0701235	Complex Analysis & Transforms	3	-	C-0700231	a,e,k
0701240	Digital Logic Systems	3	-	C-0701211	a,c,e,k
0701244	Digital Logic Systems Lab	-	1	C-0701240	a,b,c,d,g,k
0701325	Electronic Workshop	-	1	P-0701222	a,b,c,d,g,k
0701342	Microprocessors	3	-	P-0702341	a,c,e,k
0701443	Microprocessor Lab	-	1	C-0701342	a,b,c,d,g,k
0702211	Data Structures	3	-	P-0702113	a,b,c,e,k
0702202	Discrete Math	3	-	P-0700231	a, c, e, g
0702232	Object-Oriented Programming	3	-	P-0702211	a, b, c, e, k
0702233	Object-Oriented Program. Lab	-	1	C-0702232	a, b, c, e, k
0702324	Database Systems	3	-	P-0702381	a, b, c, e, k
0702325	Database Systems lab	-	1	C- 0702324	a, b, c, e, k
0702335	Probability & Statistics	3	-	P-0702201	a,e,k
0702341	Computer Organization & Architecture	3	-	P-0701242	a,b,c,g,j,k
0702342	Computer Organization & Architecture Lab	-	1	C- 0702341	a,b,e,j
0702381	Software Engineering	3	-	P-0702232	c-k
0702422	Multimedia Fundamentals	3	-	P-0702211	a,c,e,h,i,j,k
0702423	Multimedia Fundamentals lab	-	1	C- 0702422	a,c,e,h,i,j,k
0702451	Operating Systems	3	-	P-0702232	a,c,e,i,j,k
0702452	Operating Systems Lab	-	1	C-0702451	a,c,e,i,j,k
0702453	Building Web Applications	3		P- 0702324	a,b,g,l
0702454	Data Communication Systems	3	-	P-0701235	a,c,e,i,j,k
0702455	Data Communication Systems lab	1	-	C-0702454	a,c,e,i,j,k
0702456	Computer Graphics	3	-	P-0702381	a-k
0702464	Instrumentation and Control Systems	3	-	P-0701222, P-0701235	a,c,e,i,j,k
0702466	Embedded Systems	3	-	P-0701342	a,c,e,h,i,j,k
0702467	Interfacing & Maintenance lab	-	1	P-0701443	a,e,f,i,j,k
0702551	Computer Networks	3	-	P-0702451	a,c,e,j,k
0702552	Computer Networks Lab	-	1	C-0702551	a,c,e,j,k
0702492	Software Project	-	2	See Note 1	b,c,d,e,g,j,k
0702590	Internship	-	3	See Note 2	b-k
0702591	Introduction to Graduation Project	-	1	See Note 2	b,c,d,e,g,j,k
0702592	Graduation Project	-	3	P-0702591	b,c,d,e,g,j,k

Note 1: P-0702453+ Student must have successfully completed a minimum of 110 Credit Hours**.**

Note 2: P-0702467+ Student must have successfully completed a minimum of 120 Credit Hours**.**

B – Electives Courses (Total 9 credites)

Students should select Three courses from the following table.

Table III-b

Course No.	Course Name	Credits		Prerequisites /Corequisites	ILOs Covered
		Th	Lab		
0702471	Engineering Numerical Analysis	3	-	P-0700231	a,e,k
0702472	System Programming	3	-	P-0702341	a,b, c,e,k
0702474	Human-Computer Interaction	3	-	P-0702381	a,b,c,i,j,k
0702475	Advanced Computer Architecture	3		P-0702341	a,b,c,g,j,k
0702571	Networks Security	3	-	C-0702551	a,c,e,h,i,j,k
0702572	Wireless Networking	3	-	P-0702551	a,c,e,h,i,j,k
0702573	Advanced Networks	3	-	P-0702551	a,c,e,h,i,j,k
0701472	Digital Signal Processing	3	-	P-0701235	a,c,e,k
0702478	Very Large Scale Integration	3	-	P-0702342	a,c,e,k
0702476	Advanced Database Systems	3	-	P-0702325	a,c-k
0702577	Special Topics	3	-	Department Consent	a,c,e,i,j,k

Recommended Course Distribution Over the Period of Study (Semesters & Years)

First Year (Freshman)

Fall Semester				Spring Semester			
Course #	*Course Name*	*Table*	*Cr.*	*Course #*	*Course Name*	*Table*	*Cr.*
0302103	General Physics for Engineers I	II-a	3	0306102	Calculus II	II-a	3
0302113	Practical Physics for EngineersI	II-a	1	0302114	Practical Physics for Engineers II	II-a	1
0306101	Calculus I	II-a	3	0302104	General Physics for Engineers II	II-a	3
0702111	Introduction to Computer Engineering	II-a	3	0702113	Programming For Engineers	II-a	3
0702112	Introduction to Computer Enginerring Lab.	II-a	1	0702114	Programming For Engineers Lab	II-a	1
0400108	English Language Skills I	I-a	2	0700103	Engineering Drawing	II-a	1
04001xx	University Requirement	I-a	2	0400109	English Language Skills II	I-a	2
04001xx	University Requirement	I-a	2	04001xx	University Requirement	I-a	3
			17				17

Second Year (Sophomore)

Fall Semester				Spring Semester			
0700231	Engineering Mathematics	II-a	3	0701212	Electric Circuits	III-a	3
0700233	Engineering Mathematics Lab.	II-a	1	0701214	Electric Circuits Lab.	III-a	1
0701211	Intro to Circuit Theory & Elec	II-a	3	0701222	Electronics I	III-a	3
0701213	Intro to Circuit Theory & Elec Lab	II-a	1	0701224	Electronics Lab.	III-a	1
0702211	Data Structures and Algorithm Design	III-a	3	0702202	Discrete Math	III-a	3
0701240	Digital Logic Systems	III-a	3	0701244	Digital Logic Systems Lab	III-a	1
0703201	Introduction to Materials	II-a	3	0702232	Object-Oriented Programming	III-a	3
0703203	Introduction to Materials lab	II-a	1	0702233	Object-Oriented Program. Lab	III-a	1
				04001xx	University Requirement	I-a	2
			18				18

Third Year (Junior)

Fall Semester				Spring Semester			
0702381	Software Engineering	III-a	3	0702324	Database Systems	III-a	3
0700381	Engineering Management	II-a	3	0702325	Database Systems lab	III-a	1
0701235	Complex Analysis & Transforms	III-a	3	0702335	Probability & Statistics	III-a	3
0702341	Computer Organization & Architecture	III-a	3	0701342	Microprocessors	III-a	3
0702342	Computer Organization & Architecture Lab	III-a	1	0701443	Microprocessor Lab	III-a	1
0700306	Thermodynamics	II-a	3	0700209	Technical Writing & comm. skills	II-a	3
04001xx	University Requirement	I –a	2	0702464	Instrumentation and Control Systems	III-a	3
				0701325	Electronic Workshop	III-a	1
			18				18

Fourth Year (Senior)

Fall Semester				Spring Semester			
0702451	Operating Systems	III-a	3	0702466	Embedded Systems	III-a	3
0702452	Operating Systems Lab	III-a	1	0702467	Interfacing & Maintenance lab	III-a	1
0702453	Building Web applications	III-a	3	0702xxx	Department Elective	III-b	3
0702456	Computer Graphics	III-a	3	0702422	Multimedia Fundamentals	III-a	3
0702454	Data Comm. Systems	III-a	3	0702423	Multimedia Fundamentals lab	III-a	1
0702455	Data Comm. Systems Lab	III-a	1	0702492	Software Project	III-a	2
0xxxxx	Free Elective	-	3	0x001xx	University Requirement	I-b/c	3
			17				16

Fifth Year (Senior)

Fall Semester				Spring Semester			
0702551	Computer Networks	III-a	3	0702592	Project	III-a	3
0702552	Computer-Network Lab.	III-a	1	0702xxx	Department Elective	III-b	3
0702591	Introduction to the Project	III-a	1	04001xx	University Requirement	I-a	3
0702xxx	Department Elective	III-b	3	0xxxxx	Free Elective	-	3
0700xxx	Faculty Elective	II-b	3				
0x001xx	University Requirement	I-b/c	3				
0702590	Internship	III-a	3				
			17				12

Course Descriptions[1]

Faculty Compulsory Courses

♦ **0302103 General Physics for Engineers** I *(3 credits)*

Vectors, particle kinematics and dynamics, work, energy, momentum, conservation laws, collisions, rotational motion, static equilibrium, and the law of Gravity.

Prerequisite: None.

♦ **0302104 General Physics for Engineers** II**I** *(3 credits)*

Charge and matter, electric force, the electric field, Gauss's law and applications, current and resistance. Electromotive force, and circuits, the magnetic field, magnetic force on charges and currents, Ampere's and Biot-Savart laws, Inductance, and Faraday's law of induction.

Prerequisite: General Physics I 0302103.

♦ **0302113 Practical Physics for Engineers I** *(1 credit)*

12 experiments covering the topics discussed in physics 101.

Corequisite: General Physics I 0302103.

♦ **0302114 Practical Physics for Engineers II** *(1 credit)*

12 experiments covering the topics discussed in physics 102.

Corequisite: General Physics II 0302104.

♦ **0306101 Calculus I** *(3 credits)*

Limits and continuity of functions. The derivative, rate of chance, the chain rule, and related rates. Derivatives of trigonometric functions. Extreme value theorem, the mean value, and curve sketching. Anti-derivatives, the definite integral, and the fundamental theorem of calculus.

♦ **0306102 Calculus II** *(3 credits)*

Applications of definite integrals: volume, arc length, and surface area. Inverse trigonometric and transcendental functions. Techniques of integration: Integration by parts, trigonometric substitution, the method of partial fractions, etc. L'Hospitals' rule and improper integrals. Polar coordinates. Infinite sequences, upper and lower limits. Infinite series, tests for convergence, absolute and conditional convergence.

Prerequisite: Calculus I 0306101 .

♦ **0700103 Engineering Drawing** *(1 credit)*

Manual technical drawing and design: instrumental drawing, lettering, geometric constructions, sketching and shape description, multiview projection, sectional views, dimensioning. Computer

[1] Course contents are subject to change if needed.

Aided Technical Drawing and Design using AutoCAD: basic commands, settings, blocks, layers, dimensions.

♦ **0700209 Technical Writing & Communication Skills I** ***(3 credits)***

This course provides the engineering student with vital knowledge that will expand his skills in aspects of technical writing such as Technical terminology including scientific prefixes and suffixes, writing short and correct technical definitions, writing headings and titles, connecting ideas and sentences, writing a scientific paragraph and essay, writing scientific abstracts, and finally writing a technical report with a correct bibliography and citation. Oral, written and graphical communication principles are covered and exercised through the study of case histories, practical workshops and detailed assignments.

Prerequisite: English Language Skills 2 0400109.

♦ **0700231 Engineering Mathmetics** ***(3 credits)***

First and second order differential equations with applications; linear systems of differential equations. Systems of linear equations and elementary row operations, Euclidean n-space and subspaces, linear transformations and matrix representations, Gram-Schmidt orthogonalization process, determinants, eigenvectors and eigenvalues; Positive definite matrices. Computations with matrices. Unitary transforms and Hermitian matrices. QR, Singular value decomposition (SVD). Quadratic forms and convex optimization techniques.

Prerequisite: Calculus II 0306102.

♦ **0700233 Engineering Mathmetics Laboratory** ***(3 credits)***

Students will solve selected mathematical, scientific, and engineering problems using Matlab, Maple and/or Mathematica.

Corequisite: Engineering Mathmetics 0700231.

♦ **0700306 Thermodynamics** ***(3 credits)***

Definitions and basic concepts. Perfect gases, laws related to perfect gases, mixtures of perfect gases. Properties and states of simple substances. The first law of thermodynamics. Kinetic energy, potential energy, work, and heat transfers. Control volume energy analysis, conservation of mass and energy for control volume. The second law of thermodynamics, heat engines and refrigeration systems, Carnot cycles. Entropy, T & S equations.

Prerequisite: Engineering Mathmetics 0700231.

♦ **0700381 Engineering Management** ***(3 credits)***

The nature of management. Foundations of management understanding. External Environment and social responsibility. Introduction to Project Management. Development of the network and its rules. Critical path methods. Arrow Diagramming Method. Precedence Diagramming Method. Scheduling computations. Overlapping Networks. PERT Method and its statistical approach. Time-Cost Analysis. Project cost control.

Prerequisite: 3rd Year +

♦ **0701211 Introduction to Circuit Theory & Elec.** ***(3 credits)***

Basic definitions: charge, current, voltage, and power. Ohm's Law. Kirchhoff's Laws. Single loop circuits. Single node-pair circuits. Resistance and source combination. Voltage and current division. Circuit analysis techniques: Nodal, Mesh, Linearity, Superposition, Source transformation, Thevenin's and Norton theorems. Inductance and Capacitance. I-V relationships for inductors and capacitors. Insulators, conductors, and semiconductors. Intrinsic and extrinsic semiconductors. Impurities, doping, n-type and p-type semiconductors. The p-n junction diode: characteristics and applications. The Zener diode: characteristics and applications. Special purpose diodes, npn and pnp BJTs, DC biasing techniques.

Prerequisite: General Physics II 0302104.

♦ **0701213 Introduction to Circuit Theory & Elec. Lab.** ***(3credits)***

Reports and technical writing. Passive components: Resistors, capacitors, and inductors. Types, specifications and color coding schemes. Multi-output constant/variable DC power supplies: Ratings and current limitations. Digital Multi-Meters: Functions, resolution, operating ranges, and AC/DC settings. Function Generators: Signal types, duty cycle, DC offset, TTL/50Ω outputs, counter operation. Oscilloscopes: Basic theory of operation, CRTs, Start-up procedure, Phase measurement techniques. Electronic test boards: bread-boards, strip-boards, and PCBs. Mesh, Nodal and Superposition techniques. General response of an RL circuit. The Thevenin's equivalent circuit. General diode characteristics and modeling. Diode applications: rectifiers, limiters, clampers, and voltage doublers. Zener diode characteristics and applications. BJTs: characteristics, DC biasing circuits,

Corequisite: Introduction to Electrical Engineering 0701211.

♦ **0702111 Introduction to Computer Engineering** ***(3 credits)***

Introduction to computing, computer organization and architecture, bits, main & mass memory, information representation, numbering systems, Boolean operations, gates, Boolean circuits, machine language/instruction set, traditional programming concepts, procedures & functions, implementation (translation, linking, and loading), algorithm representation & discovery, flowcharting, pseudo coding, iterative structures, introduction to C programming, formatted input/output in C language, expressions in C language, one-way, two way, & multiple-way selections in C.

♦ **0702112 Introduction to Computer Engineering Laboratory** ***(1 credit)***

Computer architecture, HW & maintenance, DOS, Windows interface, files & folders, internet, email, & web search, MS office: Word/Excel/PowerPoint/Access/Visio, writing C programs with MS visual studio 6.0, C language: formatted input/output, expressions & selection statements in C.

Corequisite: Introduction to Computing 0702111.

♦ **0702113 Programming Fundamentals for Engineers** ***(3 credits)***

Revision of C language, basic types, iterative structures (loops), arrays, functions, recursion, local and global variables, pointers and arrays, pointers and functions, strings, dynamic data structures, files, introduction to OOP, classes & objects, attributes, constructors, destructors, properties, & methods, implementation using C++, other OOP concepts, graphics.

Prerequisite: Introduction to Computing 0702111.

♦ **0702114 Programming Fundamentals for Engineers Lab** ***(1 credit)***

Implementing C programs, C language basic data types, iterative structures (loops), arrays, functions, recursion, local and global variables, pointers and arrays, pointers and functions, strings, dynamic data structures, files, creating classes & objects in C++, attributes, constructors, destructors, properties, methods, graphics, demonstrating engineering and business oriented problems.

Corequisite: Programming Fundamentals for Engineers 0702113

♦ **0703201 Introduction to Materials Engineering *(3 credit)***

Mole Concept, atomic structure, periodic table, electron configuration, chemical bonding, molecular orbitals, classification of materials, the structure of crystalline solids, imperfection in solids, diffusion, general properties of engineering materials.

Prerequisite: General Physics I 0302103.

♦ **0703203 Introduction to Materials Engineering Laboratory *(1 credit)***

Investigations of various properties of materials including: mechanical, thermal, structural, chemical, electrical, and physical.

Corequisite: Introduction to Materials 0703201.

Faculty Elective Courses

- **0700382 Engineering Economy *(3 credits)***

Engineering and engineering economy. Economic and cost concepts. Interest formula derivations. Calculations of economic equivalence. Inflation and purchasing power of money. Bases for comparison of alternatives. Decision making along alternatives. Evaluating replacement alternatives. Breakeven and minimum cost analysis. Evaluation of public activities.
Corequisite: Engineering Management 0700381.

- **0700383 Quality Control *(3 credits)***

Deming philosophy and Taguchi approach in quality control. Statistical data characterization. Shewhart's Concept of Statistical Process Control. Basis for and construction of Shewhart Control Chart for Variable Data. Application of Xbar and R control Charts. Rational Sampling and Analysis of out-of-control Patterns. Process Capabilities. Tolerance Assessment. Taguchi's Loss Function. Tolerance Engineering. Statistical Thinking. X, Rm Control Charts. Exponentially Weighted Moving Average Control Charts. Cusum Control Charts and Cusum Plots. Shewhart Control Charts for Non-conformities. Bivariate Control Charts.
Corequisite: Engineering Management 0700381

- **0700385 E Business *(3 credits)***

This course outlines every aspect of e-commerce, for professional managers. It starts with an introduction to e-business, e-commerce, the Internet, the World Wide Web, and the e-business patterns, covering such topics as constructing an e-business (design, development, and management), online monetary transactions, wireless and m-commerce, e-marketing, e-publishing, e-learning, Internet security, e-customer relationship management, online industries, online banking and investment, and online career services.
Prerequisite: 3rd Year +.

Program Compulsory Courses

0701212 Electric Circuits ***(3 credits)***

RL and RC circuits. The unit step forcing function. Natural and forced response for RL and RC circuits. Parallel and series RLC circuits. The complete response of RLC circuits. Forced response to sinusoidal forcing functions. The complex forcing function. The Phasor concept. Impedance. Admittance. The sinusoidal steady-state response. Standard analysis techniques. Phasor diagrams. Response as a function of ω. Instantaneous power. Average power. Effective (RMS) values. Apparent power. Power factor. Complex power. Polyphase circuits. Single-phase three-wire systems. Complex frequency. Frequency response as a function of σ. The complex-frequency plane. Calculating natural, forced, and complete responses using the s-plane. Parallel and series resonance.

Prerequisite: Introduction to Circuit Theory & Electronics 0701211

- **0701214 Electric Circuits lab.** ***(1 credit)***

Transient response in RLC circuits. The sinusoidal steady-state response. Δ-to-Y and Y-to-Δ conversions. DC and AC current Bridges. Single stage RC and RL filters. Series RLC resonance. Computer simulations for electric circuits using Electronics Workbench, PSpice, etc.

Corequisite: Electric Circuits 0701212.

- **0701222 ElectronicsI** ***(3 credits)***

Small signal models for BJT amplifiers. The JFET: n-channel and p-channel JFETs, DC biasing techniques for JFETs, small signal models for JFET amplifiers.

Prerequisite: Introduction to Circuit Theory & Electronics 0701211

- **0701224 ElectronicsI Laboratory** ***(1 credit)***

BJT amplifiers. JFETs: characteristics, DC biasing circuits, and amplifiers.

Corequisite: ElectronicsI 0701222.

- **0701235 Complex Analysis & Engineering Transforms** ***(3 credits)***

Complex Analysis: complex numbers, analytic functions, elementary functions, complex integration, residue integration method, and conformal mapping. Fourier Analysis: periodic functions, trigonometric series, Fourier series and integrals, complex Fourier spectrum, Fourier transform and its properties. Laplace Transform: Forward and inverse Laplace transforms, linearity, applications and properties. Practical exercises using MATLAB.

Corequisite: Engineering Mathematics 0700231.

- **0701240 Digital Logic Systems** ***(3 credits)***

Boolean algebra and its laws, theorems, operations, simplification of Boolean expressions. Description the logic system behavior in an algebraic expression. Using Karnaugh Maps and Quinn-McLowski's techniques. Combinational logic networks and applications. Logic functions implementation using Multiplexers, Decoders, Read-Only Memories, and Programmable Logic Arrays. SR, JK, T, and D flip-flops. Design and analysis of synchronous sequential logic networks and applications.

Corequisite: Introduction to Circuit Theory & Electronics 0701211.

- **0701244 Digital Logic Systems Laboratory** ***(1 credit)***

  Combinational Logic circuits. Digital comparators and adders. Code conversion and multiplexers. Sequential circuit design. Counters. Sequential adders/subtractors. Shift registers. Project.

  Corequisite: Digital Logic Systems 0701240.

- **0701325 Electronics Workshop** ***(1 credit)***

  Schematics, Simulations, and PCB layouts Computer Aided Design using OrCAD, Pspice and/or Proteus etc.

  Prerequisite: ***Electronics I 0701222.***

- **0701342 Microprocessors** ***(3 credits)***

  Basic architecture of the Intel microprocessor family (XX86 and Pentium versions). Real-mode assembly language programming. Memory mapping, interfacing, and addressing. I/O interface. Programmable Peripheral Interface (PPI), Serial I/O interfacing, Interrupt Controller (PIC). DMA modules. Arithmetic Coprocessor. Bus interfacing modules.

  Prerequisite: Computer Organization & Architecture Lab 0702341

- **0701443 Microprocessor Laboratory** ***(1 credit)***

  A collection of experiments to enhance understanding microprocessor hardware bases, interfaces with real world including peripheral interfaces & programmable communication. The experiments involve machine language programming. An Intel 8088 microprocessor development kit will be used by students for "hands on" program development.

  Corequisite: Microprocessors 0701342

- **0702202 Discrete Math** ***(3 credits)***

  Introduction to networks, recursion, shortest route, Dynamic programming, Flows in networks, Linear programming, Simplex method, transportation problem, game theory, simulation.

  Prerequisite: Engineering Mathematics 0700231

- **0702211 Data Structures and Algorithm Design**

  Data representations. Algorithms design for data structures. Programming methodology. Design specifications. Structured programming. Testing strategies and algorithm analysis. Sorting and searching.

  Prerequisite: Programming Fundamentals for Engineers 0702113

- **0702232 Object-Oriented Programming** ***(3 credits)***

  Basic concepts. Classes. The inheritance concept. Polymorphism, exceptions, operator overloading, templates, visibility and dependency, patterns and frameworks. Implementation in C++/Java.

  Prerequisite: Data Structures and Algorithm Design 0702211.

♦ **0701333 Object-Oriented Program Laboratory** ***(1 credit)***

Problem solving techniques for engineering problems, primarily from the fields of electrical and computer engineering; object-oriented programming concepts; object-oriented program development, editing, compiling, linking, and debugging using the Object Oriented Programming languages.

Corequisite: Object-Oriented Programming 0702232.

♦ **0702324 Database Systems** ***(3 credits)***

This course provides an introduction to database systems including database querying, design, and programming. The course consists of three major components. The first component explains databases from a user perspective including how to query using SQL, relational algebra, and other query languages. The second component involves designing relational databases using Entity-Relationship (ER) diagrams and other modeling languages. The last part involves database programming with current technologies such as JDBC, JSP and Web technologies. Students completing the course have experience with current database technologies, and the ability to use and develop databases user interfaces, queries, forms and reports, and associated applications.
Prerequisite: Software Engineering 0702381.

♦ **0702325 Database Systems Laboratory** ***(1 credit)***

Install and configure SQL Server, Manage database files, Backup and restore databases, Manage security, Transfer data into and out of SQL Server, Automate administrative tasks, Replicate data between SQL Server instances, Maintain high availability. Creating a Database, Monitor SQL Server.

Corequisite: Database Systems 0702324

♦ **0702335 Probability & Statistics** ***(3 credits)***

Statistics: Sampling distributions, Point and interval estimation, Testing of hypothesis, Goodness of fit and contingency tables, Linear regression. *Probability Theory*: Probability, conditional probability, random variables, expected value, specific discrete and continuous distributions, e.g. binomial, Poisson, geometric, Pascal, hypergeometric, uniform, exponential and normal, Poisson process, Multidimensional random variables, Multinomial and bivariate normal distributions, Moment generating function, Law of large numbers and central limit theorem

Prerequisite: Discrete Math 0702201.

♦ **0702341 Computer Organization & Architecture** ***(3 credits)***

Assembly level machine organization, Basic organization, Control unit; instruction fetch, decode, and execution Instruction sets and types (data manipulation, control, I/O) Assembly/machine language programming Instruction formats Addressing modes I/O and interrupts. Memory system organization, Storage systems and technology, Memory hierarchy, Main memory organization and operations, Latency, cycle time, bandwidth, and interleaving, Cache memories (address mapping, replacement and store policy). I/O and communication, Input/output control methods, interrupts, Synchronization, open loop, handshaking, External storage, physical organization, and drives, Bus systems, control, direct-memory access (DMA). CPU implementation, Hardwired realization of CPU, Microprogrammed realization; formats and coding, Varieties of instruction formats, Architectural support for operating systems and compilers, Instruction pipelining, Introduction to instruction-level parallelism (ILP).

Prerequisite: Digital Logic Systems 0701242

♦ **0702342 Computer Organization & Architecture Lab** ***(1 credits)***

VLSI design packages are used to design and implement various components like ALUs, instruction decoders, register banks, processors, memories, programmable I/O devices, and ASICs.

Prerequisite: Computer Organization & Architecture 0702341

♦ **0702381 Software Engineering** ***(3 credits)***

Emergent system properties, systems engineering, legacy systems, software process models, process iteration and activities, the rational unified process, computer-aided software engineering, management activities, project planning & scheduling, risk management, functional and non-functional requirements, user & system requirements, interface specification, feasibility studies, requirement elicitation, analysis, validation and management. System models: context, behavioral, data and object model. Formal specification. Architectural design. Object-oriented design process and evolution.

Prerequisite: Object-Oriented Programming 0702232.

♦ **0702422 Multimedia Fundamentals** ***(3 credits)***

What is multimedia, overview of multimedia software tools, multimedia authoring & authoring tools, VRML, Graphics/Image data types, popular file formats, Color science, spectral sensitivity of the eye, camera systems, gamma correction, CIE chromaticity diagram, XYZ to RGB transform, Lab (CIELAB) color model, CMY color model, Color models in video: YUV, YIQ & YCbCr Color models, Types of video signals, Analog (NTSC, PAL SECAM) video and digital video (CCIR & HDTV). Digitization of sounds, Nyquist theorem, SNR, SQNR, MIDI, Quantization and Transmission of Audio: DPCM, DM , ADPCM).Lossless compression algorithms: Run-Length coding, Shanon-Fano algorithm, Huffman coding, adaptive Huffman coding, dictionary based coding arithmetic coding, Lossless Image compression.

Prerequisite: Data Structures 0702211

♦ **0702423 Multimedia Fundamentals Laboratory** ***(1 credit)***

Offering hand-on training on the latest multimedia tools & packages including graphical, animation, sound editing, video-productions, authoring. In addition, students will be trained to program in scripting languages such as XHTML, XML, and Java scripts as well as to develop multimedia-based applications in Visual BASIC, Visual C++, Java or C#.

Corequisite: Multimedia Fundamentals 0702422

♦ **0702451 Operating Systems** ***(3 credits)***

Operating System Objectives and Functions, Process definition, Process States, Operating System Control Structures, Modes of Execution, Process Creation, Process Switching, Nonprocess Kernel, Execution within User Processes, Process-Based Operating System, Unix SVR4 Process Management, Processes and Threads, Symmetric Multiprocessing, Microkernels, Thread and SMP Management of Windows, Solaris and Linux. Principles of Concurrency, Mutual Exclusion: Hardware Support, Semaphores, Monitors, Message Passing, Readers/Writers Problem. Deadlock and starvation. Memory Management Requirements, Memory Partitioning (Fixed Partitioning, Dynamic Partitioning, Buddy System, Relocation), Paging, Segmentation. Virtual memory. I/O Devices, I/O Buffering, Disk Scheduling, Raid, Disk Cache. File Management Systems, Pile, Sequential File, Indexed Sequential File, Indexed File, Direct or Hashed File, File Directories, File Sharing.

Prerequisite: Object-Oriented Program Lab 0702233.

- **0702452 Operating Systems Laboratory *(1 credit)***

  The students will explore a variety of OS mechanisms, including concurrent programming, address spaces and scheduling, virtual memory, file systems, networking by using OS simulators such OSP, Nachos, BACI or others. Windows NT: Administration, Installation, and Configuration. Management aspects such as managing of system policies, partitions, fault tolerances (RAID) and trouble-shooting tools. UNIX: Linux installation basics, general commands, the UNIX file system, shell programming, system administration including user administration, file system and disk administration, networking.

  Corequisite: Operating Systems 0702451.

- **0702453 Building Web Applications *(3 credits)***

  Introduction. Technologies of the web. JavaScript and other client-side programming within a web browser. CGI and other server-side programming with web-based application servers. Applications

  Prerequisite: Database Systems 0702324.

- **0702454 Data Communication Systems *(3 credits)***

  The course covers the principles of communication systems: Fourier and Hilbert transforms. Amplitude modulation. Angle modulation. Demodulation of FM. Noise. Potential topics include analog and digital signaling; data encoding and modulation; Shannon channel capacity; synchronous and asynchronous transmission; RS232 physical layer interface standards; FDM, TDM, and STDM multiplexing techniques; inverse multiplexing; analog and digital transmission; V series modem standards; PCM encoding, basic transmission concepts, transmission media, circuit control, line sharing techniques, physical and data link layer protocols, error detection and correction, data compression, common carrier services and data networks, and the mathematical techniques used for network design and performance analysis. Framing, queuing analysis topology optimization techniques; switching.

  Prerequisite: Probabil24ity & Statistics 0702335.

- **0702455 Data Communication Systems Lab *(3 credits)***

  Analog communication systems: AM with full carrier, AM with no carrier, DSB-SC, SSB, generation of FM, demodulation of FM, the effect of noise on FM demodulation. *Digital communication systems:* sampling, data formatting, synchronization, time-domain and frequency-domain analysis of pulse trains, PCM encoding, Interlayer Communication. Sliding Window Protocol and Go-back-N ARQ. CSMA/CD Media Access Control, Ethernet (IEEE 802.3).Token Ring Network. Transparent Bridge.

  Prerequisite: Probability & Statistics 0702453.

- **0702325 Computer Graphics *(3 credits)***

  Graphic systems. Video Display Devices. Fundamental techniques in graphics. Basic rendering and basic geometric modeling. Visualization.

  Prerequisite: Software Engineering 0702381.

♦ **0702464 Instrumentation and Control Systems** ***(3 credits)***

Instrumentation: Electronic Instrument System, Experimental Error, Sensors for Transducers (Potentiometers, Differential Transformers and Resistance Gages Capacitance Sensors, Piezoelectric Sensors, Resistance Temperature Detectors and Thermocouples). Digital Recording Systems: Conversion Processes, Digital-To-Analog Converters, Analog-To-Digital Converters. Signal Conditioning Circuits, Potentiometer Circuit, Wheatstone Bridge, Operational Amplifiers. *Control Systems*: Introduction to Control Systems, Mathematical Modeling of Dynamic Systems, Transfer Function and Impulse-Response Function, Block Diagrams, Mechanical Systems, Electrical Systems. Transient-Response Analysis, Root-Locus Analysis, Root-Locus Plots, Control Systems Design by the Root-Locus, Frequency response analysis, Bode diagrams.

Prerequisite: Electronics 0701222+Complex Analysis & Transforms 0701235

♦ **0702466 Embedded Systems** ***(3 credits)***

Embedded systems characteristics. Microprocessors versus micro controllers. Micro controller characteristics. General-purpose micro controllers. Examples of micro controller architectures. Interrupts, counters/timers, Input/output ports. Micro controller programming. Instruction set. Program development and use of assemblers. Memory maps and addressing modes. Digital to analogue and analogue to digital conversion in micro controllers. Data acquisition and distribution. Serial and parallel communications. Real-time system and its constraints. Interfacing to external devices.

Prerequisite: Microprocessors 0701342.

♦ **0702467 Interfacing & Maintenance Laboratory** ***(1 credit)***

This laboratory is divided into three practical parts: Part 1 is concerned with the hardware of the personal computer. This includes the introduction of the motherboard, the CPU, the memory, the mass-storage devices as well as the case and the power supply. Part 2 deals with the disk partitioning followed with the installation of more than one operating system, installation of office software such MS office and the usage of software tools for error debugging and repairing of the PC including the hardware and the operating system. Part 3 covers different interfacing techniques between the personal computer and external systems (automation systems) including actuators and sensors using ADC and DAC circuits, Hardware diagnosis, Bench Marking.

Corequisite: Microprocessor Lab 0701443.

♦ **0702551 Computer Networks** ***(3 credits)***

The course is an introduction to communications and networking. Topics covered include: Distinction between signal and data, and between analogue and digital data. Transmission media. Switching: Circuit-switched, packet-switched and message-switched, Network Models OSI and focusing on TCP/IP model. Connecting LANs Wired, Wireless and Virtual LANs. Multiple access protocols for Ethernet and token passing systems. Protocol hierarchies, routing of packets in networks, Internetworking, Address Mapping, Multicasting. Performance, flow and error control, and reliability.

Prerequisite: Operating Systems 0702451

- **0702551 Computer Networks Laboratory** ***(1 credit)***

The objective of this lab-based course is to gain knowledge of fundamental computer networking issues through hands-on experiments with network equipment and services. The sequence of labs start at the physical layer and progress up the protocol stack to the application layer. Topics covered are: Signal generation and analysis at the physical layer; Ethernet and WLAN performance and management; IP address planning and management; IP router generation including RIP, OSPF, BGP, MPLS protocols, TCP connection control; Stateful packet filtering; Network monitoring and management; Signaling protocols for VOIP services, and Web-based services configuration.
Experiments cover the topics of OPNET simulator which provides a very detailed modeling tool for students to build a "real" system by building a communication network in real world.

Corequisite: Computer Networks 0702551

- **0702492 Software Project** ***(2 credits)***

In this project the student is expected to use software engineering skills to design a software package which integrates the knowledge he/she gained in one or more directions.

Prerequisite: Building Web applications 0702453 +

Student must have successfully completed a minimum of 110 Credit Hours.

- **0702590 Internship** ***(3 credits)***

Each student is expected to spend at least 135 hours of training and work in a recognized engineering company or establishment. The student is required to prepare a detailed report summarizing the practical work experience both office and field work. The academic staff from the department will continuously follow up the training by making regular field visits to training locations.

Prerequisite: Interfacing & Maintenance Laboratory 0702467 +

Student must have successfully completed a minimum of 120 Credit Hours.

- **0702591 Introduction to Graduation Project** ***(1 credit)***

This course allows students to work with a faculty advisor to complete a proposed graduation project. Students are expected to gather the necessary information, scientific literature, and materials related to the project. Students are also expected to complete the initial design and implementation stage of their project. Depending on the selected project, this might involve testing hardware circuits, writing test programs, or performing any necessary computer simulations etc.

Prerequisite: Microprocessors 0701342 +

Student must have successfully completed a minimum of 120 Credit Hours.

- **0702592 Graduation Project** ***(3 credits)***

Students are expected to continue the project started in 0702591. At the end of the term, students should submit a fully documented report of their project, and perform a public presentation, open to all faculty members and any invited visitors, which includes an oral examination of their work.

Prerequisite: Introduction to the Project 0702591

Program Elective Courses

- **0702471 Engineering Numerical Analysis** ***(3 credits)***

  Floating point arithmetic. Error. Stability. Convergence. Taylor series. Iterative techniques. Curve fitting and approximation. Numerical differentiation and integration. Explicit and implicit methods.

  Prerequisite: Engineering Mathematics 0700231

- **0702472 System Programming** ***(3 credits)***

  History. Virtual machines. Data types. Control. Type checking. Run-time storage management. Finite state automata. Regular expressions, grammars and parsers, language translation systems.

  Prerequisite: Software Engineering 0702381

- **0702474 Human-Computer Interaction** ***(3 credits)***

  Principles. Mapping. Constraints. Information visualization. Modeling. Interaction. Window management. Help systems. Evaluation techniques. Computer-supported collaborative work.

  Prerequisite: Software Engineering 0702381

- **0702475 Advanced Computer Architecture** ***(3 credits)***

  Conventional uni-processors, pipelined processors, parallel processors & multiprocessors. Memory & cache structures .Multiprocessor algorithms & synchronization. Special-purpose architectures.

  Prerequisite: Computer Architecture 0702341

- **0702571 Network Security** ***(3 credits)***

  This course explains Security protocols, authentication protocols, data integrity, digital signatures, intrusion detection, key management and distribution, viruses and other malicious codes, information flow, mobile code and agent security. Cryptographic algorithms: Secret Key Encryption (DES), Public Key Encryption (RSA), Message Digest Algorithm (MD5); Attacks and countermeasures: Packet sniffing, Spoofing and denial of service; Application layer security: HTTPS, secure email; Transport layer security: TLS, SSL; Network layer security: IP security (IPSec), AH protocol, ESP protocol; access control and Firewalls: Filter-based firewalls, Proxy-based firewalls; wireless networks security, security in IEEE 802.11, WEP protocol, EAP protocol.

  Corequisite: Computer Networks 0702551.

- **0702572 Wireless Networking** ***(3 credits)***

  This course covers fundamental techniques in design and operation of wireless networks: Wireless LAN, the IEEE802.11 standard, Ad Hoc nets, Bluetooth technology, GSM and also wireless sensor networks. existing systems for satellite communication, mobile telephony, local wireless networks

and mobile IP. also acquire an overview of current technologies and problems such as: Antennas, wave propagation, power control, multipath, propagation, fading, frequency hopping.

Prerequisite: Computer Networks 0702551.

- **0702573 Advanced Networks** ***(3 credits)***

Circuit Switching and Packet Switching, Routing in Switched Networks, Transport Protocols Congestion Control in Data Networks, Cellular Wireless Networks, Wireless LANs, Internetwork Operation, SONET/SDH, Frame Relay and ATM, Application Layer, Multimedia, Network Security, Sockets Programming.

Prerequisite: Computer Networks 0702551.

- **0701472 Digital Signal Processing** ***(3 credits)***

Signals, systems and signal processing. Discrete-time signals. Frequency analysis. Z-Transform. Analysis and design of discrete systems. Sampling. FIR, IIR structures. Discrete Fourier Transform.

Prerequisite: Complex Analysis & Transforms 0701235

- **0702478 Very Large Scale Integration Design** ***(3 credits)***

Introduction to Microelectronics. Semiconductor Physics and IC Technology. Metal-Oxide Semiconductor (MOS) Transistors. MOS Inverters and Gate Circuits. Bipolar Transistor Inverter. Bipolar Gate Circuits. Regenerative Logic Circuits. Semiconductor Memories. VLSI Systems: Design, Fabrication and Layout.

Prerequisite: ***Computer Architecture 0702341 .***

- **0702577 Special Topics** ***(3 credits)***

This course has a very flexible and varied outline that covers various topics depending on the research interests of the academic instructor teaching it.
Prerequisite: Department Consent.